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REVIEW ARTICLE

Constructed wetlands as sustainable solutions for textile effluent treatment: a review

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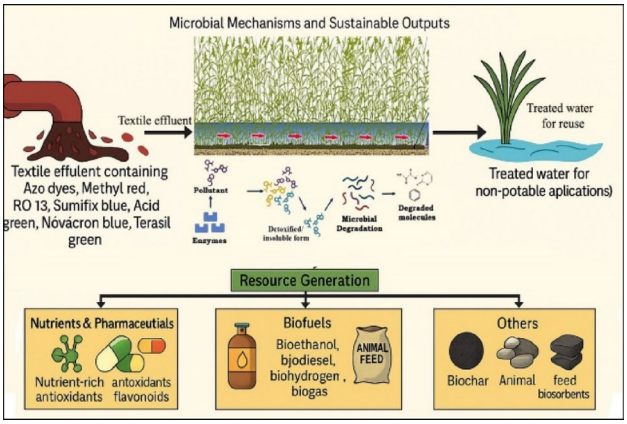
Abstract

Constructed Wetlands (CWs) have emerged as a promising technology for the effective remediation of wastewater, including textile effluent, and provide an alternative water source for various activities. Several bacterial populations are critical in degrading and decolorizing noxious dyes within CWs. However, enzymes are considered highly efficient biological catalysts utilized to speed up degradation reactions within CWs for the removal of different dyes from textile wastewater. The obtained macrophytic biomass acts as a raw material to develop different bioresources such as biofertilizers, biofuels, bioenergy, animal feed, biochar, recovery of various nutrients and pharmaceutical compounds, etc. Several macrophytes utilized in CWs have great nutritional and pharmaceutical values. Further, the production of biofuels from biomass feedstock has been extensively investigated to develop an alternative fuel for reduced dependence on fossil fuels. Additionally, strong, robust, and air-dried stem strands of macrophytes obtained during the treatment process can be utilized as an alternative structural resource. The resultant sludge and other solid, biosolid, and semi-solid materials may produce heat or electricity, biogas, or be transformed into organic manure or fertilizers. This review highlights the potential of CWs for textile wastewater treatment and the generation of various value-added products for the circular economy. The role and potential of microbial populations and their enzymes during the treatment process are also discussed briefly.

Graphical abstract



Keywords Wastewater treatment · Nature-based solutions · Bioresource · Circular economy · Sustainability

Introduction

Rapid population growth and urbanization have led to 56% of the world's population (4.4 billion people) residing in cities, a figure expected to double by 2050 (Kumar et al.

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2022a, b). This surge intensifies natural resource demand, with global material consumption projected to reach 90 billion tons by 2050 (Swilling et al. 2018). Despite the Sustainable Development Goal (SDGs-6) emphasis on clean water access and ecosystem protection, 80% of wastewater is discharged untreated worldwide, posing severe environmental and public health risks. The textile industry is a major contributor to this crisis, with dyeing alone generating 80% of its wastewater (Keskin et al. 2021). Producing 1 kg of textiles consumes 100 L of water (Ali 2010), while effluent contains 5–35% of the 10,000 chemicals used in manufacturing (Verma et al. 2012). Azo dyes, prevalent due to their durability, are particularly toxic, harming aquatic ecosystems and human health through bioaccumulation (de Campos Ventura-Camargo and Marin-Morales 2013). Textile wastewater's variable pH (5–10), high salinity, and organic load disrupt aquatic life by blocking light, reducing dissolved oxygen, and causing mutagenic effects (Samsami et al. 2020; Mehra et al. 2021). Further, they suppress the plant growth and development and CO_2 assimilation, toxic dyes and their derivatives encourage aquatic toxicity, including mutagenicity and carcinogenicity (Mehra et al. 2021). Dyes pose huge ecological pollution and hazards for all living beings due to their extensive application and carcinogenic and mutagenic effects (Keskin et al. 2021; Sharma and Malaviya 2022). Several other wastes that come from textile manufacturing, such as plastics, polyesters, and yarns, etc. are accountable for the pollution of agricultural lands and local habitats, particularly in developing nations (Slama et al. 2021; Pratap et al. 2023, 2025).

Although numerous physicochemical treatment methods have been developed, such as coagulation, flocculation, oxidation, and membrane-based filtration, etc., have been applied for the management of textile effluent as explained in Fig. 1 (Mirza et al. 2020). These are often criticized for their high operational costs, chemical dependency, generation of secondary waste, and energy demands. These limitations necessitate the development of alternative, eco-efficient, and sustainable treatment strategies that are aligned with both environmental and economic priorities. In recent years, the concept of nature-based solution (NBS) has gained traction as a strategic approach to address complex environmental challenges, including water pollution, climate change, and biodiversity loss, through the sustainable use of natural processes. Among various NBS technologies, constructed wetlands CWs have emerged as one of the most robust and scalable alternatives for treating industrial wastewater, especially from the textile sector, which is notorious for releasing large volumes of complex and persistent pollutants. CWs have gained increasing global attention as a NBS that mimics natural processes for wastewater treatment. Their low energy requirements, ease of operation, adaptability, and ability to support biodiversity make them highly attractive for both urban and rural wastewater management, particularly in developing countries.

CWs offer a highly adaptable framework that can be customized based on the wastewater load, available land, local climate, and target contaminants, making them particularly suitable for decentralized treatment in resource-constrained settings. Their ability to support diverse biological communities, including bacteria, fungi, and macrophytes, enables effective biotransformation of pollutants such as dyes, heavy metals, nutrients, and endocrine-disrupting compounds (Vymazal 2013; Kumar and Dutta 2019).

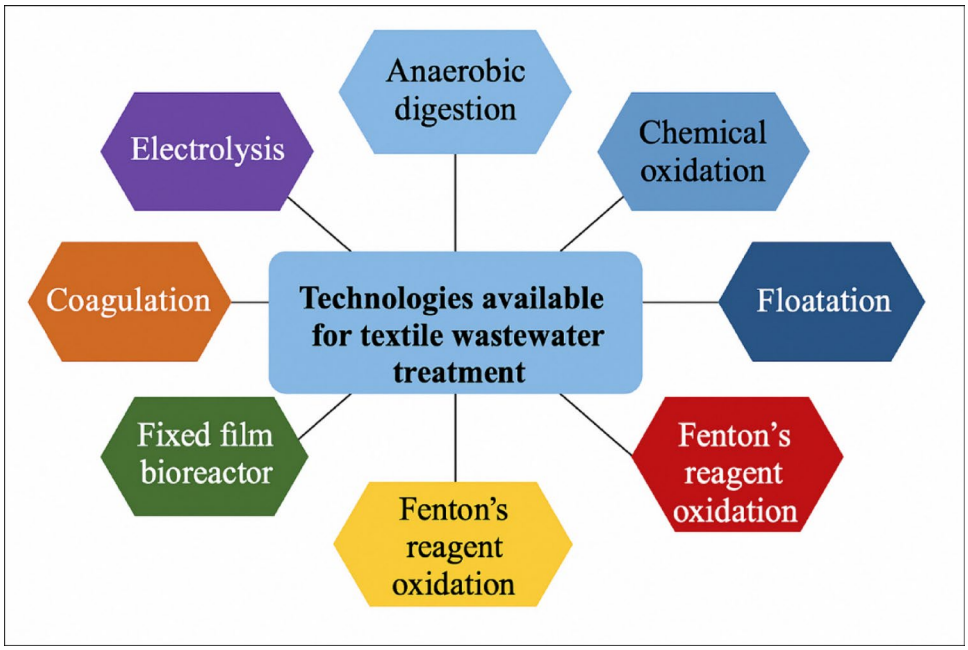

Fig. 1 Conventional technologies of textile wastewater treatment

These systems utilize physical (sedimentation, filtration), chemical (adsorption, photodegradation), and biological (microbial and plant uptake) mechanisms, providing a multi-barrier approach to wastewater treatment (Liang et al. 2017; Shindhal et al. 2021). Furthermore, the suitability of CWs for textile effluent treatment lies not only in their pollutant removal efficiency but also in their resilience, low operational cost, and minimal energy footprint (Yaseen and Scholz 2019). Unlike conventional treatment systems that often require constant monitoring and high energy inputs, CWs operate with passive mechanisms and require limited technical expertise, making them an attractive solution in both developed and developing countries (Zheng et al. 2022; Hdidou et al. 2021). CWs demonstrate immense potential for resource recovery and waste valorization. The biomass generated from aquatic macrophytes, which absorb and accumulate nutrients and organics during wastewater treatment, can be repurposed into a range of bioresources, including biofertilizers and organic compost, enriched with nitrogen, phosphorus, and potassium (Saliu and Oladoja 2020; Li et al. 2021). Bioenergy sources such as biogas, bioethanol, and biodiesel through anaerobic digestion or pyrolysis (Rezania et al. 2020). Such multipurpose utility of CWs positions them as a unique solution that supports the "waste-to-wealth" concept and aligns with global efforts to implement sustainable water management and circularity in the industrial sector. Moreover, hybrid CW designs which combine horizontal and vertical flow systems have demonstrated superior performance in removing recalcitrant compounds from textile effluent under varied redox conditions (Saeed and Sun 2013; Kumar and Dutta 2019; Benny and Chakraborty 2023). Despite the growing body of literature on CWs, a critical gap exists in evaluating their multifaceted roles beyond pollutant removal. Most previous reviews have focused either on the design, operation, and treatment efficiency of CWs or have analyzed specific pollutant removal mechanisms (Hassan et al. 2021). However, very few studies have provided an integrated view of CWs as platforms for both wastewater remediation and the generation of value-added bioresources, which is vital in advancing circular economy models.

The primary objective of this review is to provide an integrated and multidisciplinary perspective on the role of CWs in textile effluent treatment, with an emphasis on their dual functionality as both pollutant removal systems and platforms for resource recovery. While many studies have examined the performance of CWs in degrading dyes and industrial pollutants, there remains a critical gap in assessing their broader contributions to sustainable development and circular economy frameworks (Tisserant et al. 2017; Winans et al. 2017). This review addresses that gap by (i) analyzing the role of microbial populations and their enzymatic mechanisms in textile dye degradation within CWs, (ii) highlighting the applicability and efficiency of CWs in removing diverse textile pollutants, and (iii) exploring the valorization potential of the biomass generated through CWs for recovery of nutrients, biofuels, biofertilizers, pharmaceuticals, biochar, and other high-value bioresources. By focusing on these aspects, the review underscores CWs not only as eco-technological solutions for wastewater remediation but also as enablers of resource recovery and sustainability. This comprehensive approach aims to inform researchers, policy-makers, and practitioners about the transformative potential of CWs in promoting low-cost, decentralized, and nature-based strategies for industrial wastewater management.

Application of CWs for textile wastewater treatment

CWs are designed to exploit the natural ongoing functions of vegetation, substrate materials, and the connected microbial populations for the treatment of several types of wastewater, including textile wastewater (Jegatheesan et al. 2016; Kumar and Dutta 2019; Zheng et al. 2022). They combine the use of physicochemical and biological interactions to remove/ degrade contaminants from wastewater (Valipour and Ahn et al., 2016). They are classified according to the following criteria (Hassan et al. 2021)-

i. Based upon hydrology they could be either surface flow or sub-surface flow.
ii. Based upon vegetation they can be submerged, free-floating, or emergent.
iii. Based on the flow path they could be either horizontal or vertical.

CWs can be implemented either as standalone systems or in combination with multiple types to enhance treatment efficiency. The core components of CWs include substrates such as coarse gravel, sand, and soil; filler materials like activated charcoal and limestone; and various macrophytes. Commonly used macrophyte species in CWs for treating different types of wastewater include *Typha spp.*, *Scirpus spp.*, *Phragmites spp.*, *Zizania caduciflora*, *Juncus spp.*, *Glyceria maxima*, *Cyperus spp.*, *Canna indica*, *and Pistia stratiotes* (Vymazal 2013). Macrophytes play a critical role in the functioning of CWs (Ji et al. 2022). They directly assimilate nutrients and accumulate pollutants in their tissues. Their roots offer adsorption sites for contaminants and surfaces for microbial colonization. Additionally, macrophytes transport oxygen through their stems and leaves into the rhizosphere, enhancing aerobic conditions, maintaining

hydraulic conductivity, and releasing exudates that act as biochemical catalysts, thereby supporting the overall integrity and performance of the wetland system (Liang et al. 2017; Nan et al. 2020). The effectiveness of CWs largely depends on the characteristics of the macrophytes used, including their nutrient uptake capacity, oxygen-release potential, association with microbial communities, suitability as attachment surfaces, and filtration abilities (Valipour and Ahn 2016).

CWs have demonstrated high efficiency in remediating wastewater containing azo dyes (Sharma and Malaviya 2022). For instance, Ong et al. (2010) evaluated vertical flow CWs planted with macrophytes for the treatment of synthetic wastewater containing acid orange dye (50–100 mg/L). Their findings revealed excellent removal efficiencies: color (96–98%), chemical oxygen demand (COD) (74–82%), total nitrogen (44–64%), and ammonium (24–48%) using gravel as the substrate, after a hydraulic retention time (HRT) of 3–5 days. Tee et al. (2015) designed horizontal flow CWs vegetated with *Typha latifolia* and used a media blend of gravel and rice husk to treat domestic wastewater spiked with acid orange 7 dye (300 mg/L). They reported that a baffled CW configuration achieved 69–100% color removal, significantly outperforming traditional CW systems (30–73%) within 2–5 days of HRT. Similarly, Sethulekshmi and Chakraborty (2021) developed a vegetated baffled CW using gravel to treat synthetic wastewater containing reactive red dye. Their system achieved 48–73% color removal and 65–85% COD reduction at an HRT of 3–7 days.

It has been widely acknowledged that treatment technologies capable of facilitating both aerobic and anaerobic conditions are most effective in degrading azo dyes. In this regard, hybrid CW systems combining vertical-flow (VF) and horizontal-flow (HF) configurations are particularly suitable and have been adopted for treating various wastewaters, including those from the textile industry, across several countries (Saeed et al. 2012; Tee et al. 2015; Kumar et al. 2022a, b). Interestingly, the decolorization of azo dyes under aerobic conditions is significantly less efficient compared to anoxic or anaerobic settings (Kapoor et al. 2021), largely due to the electron-deficient nature of azo dyes, which makes them less susceptible to oxidative metabolism (Popli and Patel 2015). Studies have also shown that horizontal-flow CWs (HFCWs) exhibit greater color removal than vertical-flow CWs (VFCWs), mainly because of the dominant anaerobic microenvironments within horizontal beds (Saeed and Khan 2019; Benny and Chakraborty 2023). Saeed and Sun (2013) applied a hybrid CW system for textile effluent treatment and reported color removal efficiencies of 61–79% in HFCWs under anaerobic conditions and 37–61% in VFCWs under artificially aerated conditions. In another study, Saeed and Khan (2019) tested a vertical-vertical-horizontal hybrid configuration for treating mixed industrial effluent, finding color removal efficiencies of only 2–20% in the vertical flow stages compared to 81–90% in the horizontal stage. To better understand the mechanisms of dye removal, Benny and Chakraborty (2023) conducted batch experiments using autoclaved and non-autoclaved substrates in CWs. Their findings indicated that color removal was substantially higher (59 ± 5%) in non-autoclaved materials compared to autoclaved ones (9 ± 3%) at 3 days HRT, suggesting a significant role of microbial degradation in the decolorization process. The performance of CWs for the treatment of dyeing wastewater based on previous studies throughout the globe has been elucidated in Table 1.

Role of bacterial populations and their enzymes in decolorization and degradation of textile dyes within CWs

Enzymes are highly efficient biological catalysts widely used in CWs to accelerate the degradation of various dyes present in textile wastewater (Kumar et al. 2020; Varjani et al. 2020). A range of enzymes such as lignin peroxidase, manganese peroxidase (MnPs), laccase, azo-reductase, riboflavin reductase, aminopyrine N-demethylase, micro-peroxidase-11, tyrosinase, and polyphenol oxidase have proven effective in decolorizing different types of dyes. Additionally, redox systems like Dichlorophenolindophenol–Nicotinamide adenine dinucleotide hydrogen (DCIP–NADH) are commonly employed as electron carriers or indicators in assays evaluating enzymatic dye decolorization activity. (Imran et al. 2015; Mishra et al. 2019; Wong et al. 2019). One of the key steps in microbial decolorization of azo dyes is the reduction of azo bonds (-N = N-), which serve as the chromophore group. This process can occur under both aerobic and anaerobic conditions. However, the presence of external organic compounds such as starch, glucose, acetate, or ethanol is essential, as they act as electron donors required for breaking azo bonds (Popli and Patel 2015). Enzymes from the oxidoreductase family, including peroxidases, reductases, and laccases, are particularly effective in degrading dye-contaminated wastewater from various industries (Singh et al. 2015). Laccase, in particular, can oxidize a broad spectrum of compounds, including mono- and di-phenols and their derivatives containing amino, carboxy, hydroxy, methoxy, or sulfo groups (Legerská et al. 2016). These enzymes facilitate dye removal either by precipitating dye molecules or by converting them into simpler, easily removable compounds (Routoula and Patwardhan 2020). The simple figure of bacterial enzymatic degradation in constructed wetland has been shown in Fig. 3.

Table 1 The performance of CWs for the exclusion of various contaminants, including color from dyeing wastewater, throughout the world

Sl. No.	Type of CWs	Name of plants	Dye	Dye concentration (mg/L)	Decolorization time	Color removal%	BOD removal %	COD removal %	References
1.	HFCW	*Phragmites australis* and *Typha domingensis*	Acid Red 18	–	24 and 48 h	–	–	86	Hussein (2023)
2.	VFCW	*Canna Indica*	Methylene Blue dye (MB)	50	–	85	–	–	Jayalakshmi et al. (2023)
3.	VFCW	*Eichhornia crassipes*	Remazol Yellow FG	200	1 day	98.2	–	91	Teoh et al. (2023)
4.	HFCW	*Salvinia molest Aster amellus*, *Glandularia pulchella*, and *Zinnia angustifolia*	Acid Red 18, Acid Yellow 23, Reactive Black 5 and Yellow 84	–	–	–	–	–	Khan et al. (2022)
5.	VFCW	*Canna indica*	–	–	–	98.24	91	87	Karungamye (2022)
6.	CW-MFCs)	–	Methyl orange	–	–	98	–	85.29	Sonu et al. (2022)
	–	*Pistia stratiotes*	–	–	–	–	51.17	58.03	Mustafa and Hayder (2021)
	–	*Nymphaea spontanea*, *Nelumbo nucifera*, *Azolla filiculoide*, *Pistia stratiotes*	Mauveine, Brilliant Blue Green	–	36 h	94	69	73	Kaushal and Mahajan, (2021)
7.	VFCW	*Phragmites australis and Typha domingensis*	Congo red dye	10–40	1–5 day	93.1	–	99.34–56.5	Bedah and Faisal (2020)
		Phragmites australis	Azo-Dye	2.5 to 150	27 days	–	–	–	Riva et al. (2019)
8.	HFCW	*Typha domingensis*	Azo dye	10, 15, 20, and 25	17 days,	92	64–68	56 and 68	Haddaji et al. (2019)
9.	HFCW	*P. australis*	–	–	–	–	91.05	87.1	Hussain et al. (2019)
10.	HFCW	*L. fusca*	–	–	–	–	77.89	85.8	Hussain et al. (2018)
11.	VFCW	*F. dichotoma and A. baccifera*	Methyl orange	50	60 h	91 and 89	77	72	Kadam et al. (2018)
12.	HFCW	*Chrysopogon zizanioides and Typha angustifolia*	Reactive red dye	150	60-h	89 and 87	75	75	Kadam et al. (2018).
		Eichhornia crassipes	–	97	–	90, 88, 92, and 90	75	81.4	Priya and Selvan (2017)
13.	HFCW	*Typha angustifolia*, *Paspalum scrobiculatum*	Congo Red	100	96 h	62, 59 and 76	68, 63 and 75	65, 63 and 70	Chandanshive et al. (2017)
14.	VFCW	*Phragmites australis*	Basic Red 46 and Acid Blue 113	7 and 208	48 and 96 h	85 and 100	–	–	Hussein and Scholz (2017)
15.	VFCW	*Phragmites australis*	Acid Blue 113 and Basic Red 46	7 and 215	48 h and 96 h	71, 68 and 80	–	50, 59 and 67	Hussein and Scholz (2017)
16.	VFCW	*Typha angustifolia and Canna indica*	Azo-dye	1000	3.75 days	94.29,	94–99	57–63%	Dogdu and Yalcuk (2016)
17.	VFCW	*Canna indica*, *Typha angustifolia*	Vat dyes commercial indigo dye	1000	3.75 d	97	–	70	Dogdu and Yalcuk (2016)
18.	VFCW	*Dracaena*	Methyl yellow as dye	1000	–	75.1	–	50.5–72.3	Dogdu and Yalcuk, (2016)
19.	VFCW	*Phragmites australis*	Acid Orange 7, Reactive Red 141	130–170	6 days	86 and 75	66	75 and 90	Khandare and Govindwa, (2015)
20.	HFCW	*(Typha latifolia*	Azo dye, Acid Orange 7	300	5, 3 and 2 days	100, 83 and 69	–	–	Tee et al. (2015)
21.	VFCW	*Phragmites sp.*	Azo dyes (Direct Red 81)	–	7 days	96 − 89	–	90.0	Ferreira et al. (2014)

Table 1 (continued)

Sl. No.	Type of CWs	Name of plants	Dye	Dye concentration (mg/L)	Decolorization time	Color removal%	BOD removal %	COD removal %	References
22.	Surface flow	*Eichhornia crassipes*	–	–	5 days		68–78.26	55.6–87.17	AS et al. (2012)
		Canna indica	Methyl orange, Azo dye	500–1500	3 to 6 days	97	–	–	Yadav et al. (2012)
		Echinodorus cordifolius L.	Ruthenium Red 2	20	7 days	96	–	–	Noonpui and Thiravetyan (2011)
23.	VFCW	*Typha angustifolia*	Reactive azo dyes	–	30 days,	60	–	82.5	Nilratnisakorn et al. (2009)
24.	VFCW	*Phragmites sp.*	Acid Orange 7	–	5 days	63.8	–	63.8	Nilratnisakorn et al. (2009)

Genome mining of *Pleurotus ostreatus* has identified 34 putative carboxylesterase and 53 putative lipase genes potentially involved in dye decolorization (Dwivedi and Tomar 2018). Among oxidoreductases, MnPs, a group of heme-containing glycoproteins, were initially discovered in *Phanerochaete chrysosporium* and are known to degrade a variety of hazardous pollutants, including dyes, antibiotics, phenolic compounds, and polycyclic aromatic hydrocarbons. Their recyclability and cost-effectiveness make MnPs a promising tool for dye removal in wastewater treatment (Chang et al. 2021). The role of bacterial populations and their enzymes in decolorization and degradation of dyes has been provided in Tables 2 and 3.

Potential resource recovery options for the circular economy

CWs emerged as a promising technology offering efficient dye removal, recovering nutrients and providing an alternative water source for various activities such as agriculture, gardening, industrial reuse, etc. within the circular economy context (Hdidou et al. 2021). Further, the generation of bioenergy from biomass developed to decrease dependence on fossil fuels has the potential to supply about 10%, or up to 50 EJ, of global energy by 2035 (Avellán and Gremillion et al. 2019). However, other crucial benefits of CWs for wastewater treatment are their ability to offset greenhouse gas emissions and to develop bioenergy. CWs are considered net emitters of methane (CH_4), but a sink of carbon dioxide (CO_2) via the accumulation of organics and are net greenhouse gas sinks (Avellán & Gremillion et al. 2019). Several possible resource recovery options during the textile wastewater treatment through CWs are explained in Fig. 2.

Nutrient recovery

The wastewater originating from various industries such as agriculture sectors, food, municipal and household use are regarded as nutrient-rich which may serve as a valuable option to recover resources. The nutrients in wastewater generally comprise phosphorus and nitrogen, vital for the growth of macrophytes (Yan et al. 2018; Saliu and Oladoja 2020). It is reported that approximately 15– 20% of phosphorus supply can be attained via recovery from municipal wastewater worldwide (Cainglet et al. 2020; Mustafa and Hayder 2021). However, the recovery of nitrogen is measured by an energy-efficient approach utilized in fertilizer production. Readily available nitrogen can be used directly in fertilizer-developing procedures, thus minimizing the usage of resources otherwise required for fixing atmospheric nitrogen (N_2) (Rahimi et al. 2020). Furthermore, sewage contains high nutrients, where approximately 75% of nitrogen and 50– 80% of phosphorus come from washrooms (Saliu and Oladoja 2020). This also comprises certain micronutrients required in trace amounts for the growth of plants. These are Fe (iron) for numerous enzymatic responses, Mn (manganese) for several plant metabolic actions, B (boron) for the building of cells, Zn (zinc) for chlorophyll development and Ni (nickel) for the metabolism of nitrogen (Ahmad et al. 2009; Alejandro et al. 2020; Schmidt et al. 2020). The nutrient concentrations in wastewater provide huge opportunities for nutrient recovery using aquatic macrophytes within CWs and utilizing obtained biomass for the development of useful goods, hence preventing wastage and protecting the ecosystem. All these efforts are considered a vital part of the circular economy strategy, in which the main objective is to reduce or minimize waste by proper utilization of resources.

Table 2 Bacterial strains and their enzymes involved in the degradation of different dyes

Sl. No.	Bacterial strains	Involved enzymes	Name of the dye for degradation	References
1.	*Alishewanella sp.* *Pseudomonas nitro-reducers* *Vibrio logei*	Azo-reductase, Laccase, Lignin peroxidase (LiP)	Sumifex Tourqi blue Methyl Red	Mishra et al. (2022)
2.	*Alcaligenes faecalis* *Pseudomonas aeruginosa* *Lysinibacillus sphaericus*	Oxidase, azo-reductase Lipoxygenase enzyme Azo-reductase and laccase	Reactive Orange 13 (RO 13) Cleave azo bond Azo dye	Saravanan et al. (2021)
3.	*Azospirillum lipoferum* *Marinomonas mediterranea* *Pseudomonas desmolyticum* *Pseudomonas syringae* *Pseudomonas chrysosporium*	Laccase, Peroxidase	Direct Blue 6, Green HE4B Red HE7B, Anthraquinone	Indrani (2020)
4.	*Bacillus* *Enterococcus* *Lysinibacillus* *Pseudomonas* *Psychrobacter* *Shewanella*	Azo-reductase	Azo dyes	Imran et al. (2015)
5.	*Bacillus subtilis*, *Bacillus megaterium*, *Erysipelothrix* *Amphibacillus xylanus.* *Enterobacteriaceae*, *Micrococcus sp.*, *Pseudomonas sp.*, *Lactobacillus sp*, *Staphylococcus sp.*, *Alcaligenes sp.*, *and Aeromonas sp.*	Azo-reductase, Laccase, Oxidases, Peroxidase, Lignin peroxidase (LiP), Oxidoreductase	Anthraquinone, Cibacron red FN-R, Acid Green (AG) 25, Novacron blue, navy, orange and yellow, Terasil green, Reactive Red, Acid Red (AR) 18, Engineered, Remazol	Barathi and Arulselvi (2015)
6.	*Momordica charantia*	Azo-reductases, Laccases Peroxidases	Azo dyes	Rauf and Ashraf (2012)

Table 3 Degradation potential of several enzymes for various dyes as reported earlier

Sl. No.	Enzymes involved	Dyes	Dye degradation (%)	References
1.	Azo-reductase	Sumifex Tourqi blue	83%	Mishra et al. (2022)
		Reactive red 195	99.11% 100%	Mishra et al. (2022)
		Reactive red 141	– –	Parmar and Shukla (2018)
		Azo dye		Singh et al. (2015);
		Reactive Black 5	93%	Mishra et al. (2022)
		Methyl Red	100%	Bharathi et al. (2022) Khan et al. (2013)
2.	Oxidative (LiP)	Azure B	Up to 96%	Mishra et al. (2022)
		Basic red 9 dye	91.73%	Saravanan et al. (2021)
3.	Horseradish peroxidase (HRP)	acid Violet 109/30	(70 − 90%)	Routoula and Patwardhan (2020)
4.	Tyrosinases	Direct Blue 6	98%	Parmar and Shukla (2018)
5.	Peroxidase	Remazol Brilliant Blue R	57.6%	Gowri et al. (2014) Khan et al. (2013)
6.	Lignin peroxidase	Reactive Blue 59		Gowri et al. (2014)
7.	LiP, tyrosinase	Reactive blue 25	100%	Kaushik and Malik (2009)
8.	Cellobiose dehydrogenase	Acid blue 62	99%	Kaushik and Malik (2009)
9.	Laccase	Crystal Violet	100%	Mishra et al. (2022) Bharathi et al. (2022)
		Reactive Blue 4	97%	Saravanan et al. (2021)
		Methylene blue dye Indigo carmine,	38–79%	Sarkar et al. (2017)
		Congo red	99, 98, and 60%	Singh et al. (2015)
		Malachite Green	96.91%	Khan et al. (2013)

Extraction of valuable nutritional and pharmaceutical chemicals

Macrophytes, particularly *Eichhornia crassipes* (water hyacinth), are emerging as sustainable sources of bioactive compounds with high pharmaceutical and nutritional value. For instance, Adelodun et al. (2020) extracted six major organic compounds from the stem of *E. crassipes*: lauric acid (dodecanoic acid), oleic acid, 9,12-octadecadienoic acid (linoleic acid), chloroethyl ester, palmitic acid (n-hexadecanoic acid), and cis-vaccenic acid. Oleic acid, a monounsaturated omega-9 fatty acid, is a principal component of olive oil and was found to be the most abundant compound in the extract (~35.5%) (Adelodun et al. 2020). It has demonstrated antihypertensive and cardioprotective properties, and diets rich in oleic acid are linked to reduced risk of breast cancer (Jude et al. 2006). Lauric acid, found naturally in coconut and palm kernel oil, is well known for its

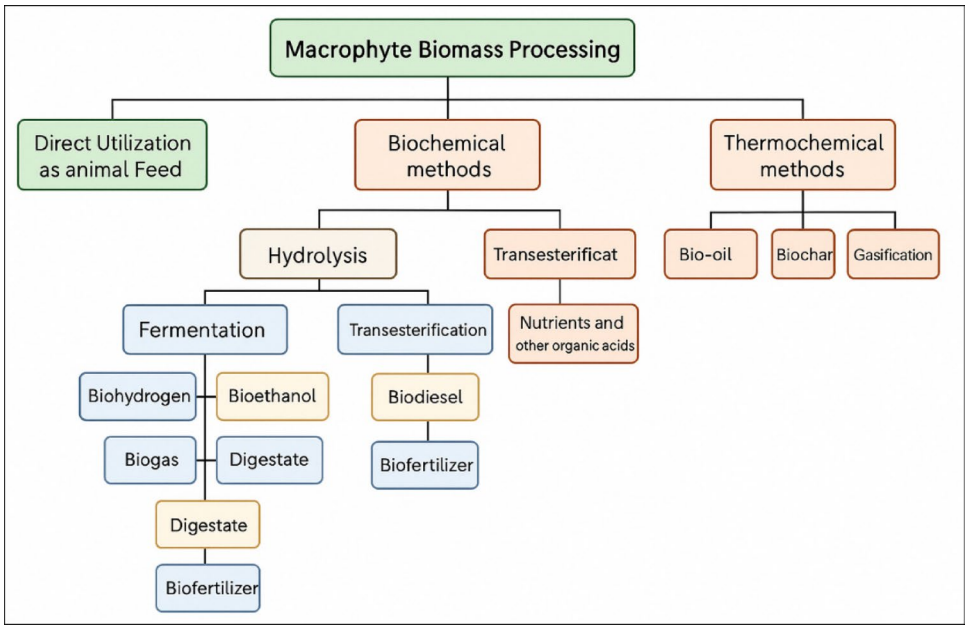

Fig. 2 Possible macrophytes biomass utilization employing various methods for the generation of valuable bioresources. (Modified from Kurniawan et al. 2021)

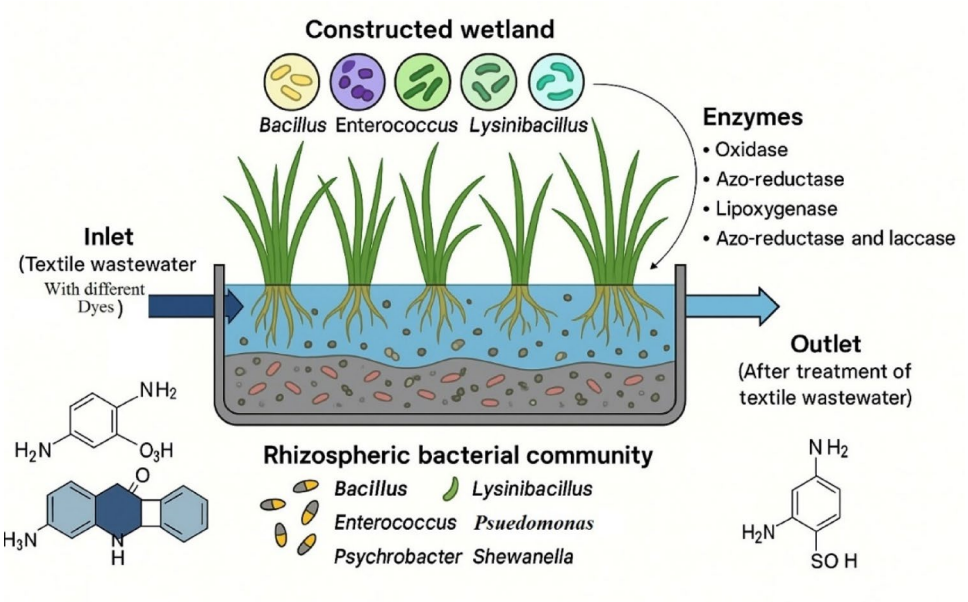

Fig. 3 Microbial mechanisms in constructed wetlands: Rhizospheric bacterial communities and their enzymes facilitate textile dye decolorization and degradation

antimicrobial action against bacteria, fungi, and viruses (Nitbani et al. 2022). It is widely used in the soap and cosmetic industry and forms part of antimicrobial drug formulations. 9,12-Octadecadienoic acid (linoleic acid) is an essential omega-6 polyunsaturated fatty acid with anti-inflammatory and hypocholesterolemic effects. It is used in nutraceutical formulations and dietary supplements due to its role in cholesterol metabolism and cardiovascular protection (Nur Mahendra et al. 2023). Cis-vaccenic acid, a positional isomer of oleic acid found in ruminant fats, has been reported to modulate immune responses and reduce cancer cell proliferation in vitro (Naim and Ahamad 2024). Its therapeutic potential in inflammatory and cardiovascular diseases has been increasingly recognized. Palmitic acid, a saturated fatty acid prevalent in plant and animal fats, is extensively utilized in cosmetic, soap, and industrial applications. Though high dietary intake is linked with adverse metabolic effects, palmitic acid is also essential for membrane lipid synthesis and protein palmitoylation (Murru et al. 2022). Chloroethyl esters, while less studied in nutritional applications, are known intermediates in the synthesis of pharmaceuticals and pesticides and are used industrially in varnishes and paints. However, their presence in biological systems is usually monitored due to their potential toxicity (Peng and Shahidi 2024). Thus, *Eichhornia crassipes*, often viewed as an invasive weed, holds promise as a low-cost biomass for

extracting value-added compounds with significant pharmaceutical and industrial applications. The integration of such natural products into circular bioeconomy models can aid in environmental management while supporting pharmaceutical innovation.

Macrophytic biomass utilization

CWs planted with macrophytes such as *Eichhornia crassipes*, *Typha latifolia*, and *Phragmites australis* are highly effective for the treatment of wastewater containing elevated levels of organic matter and nutrients. These systems not only provide eco-technological purification but also result in the rapid accumulation of large quantities of biomass, which poses a challenge for post-treatment management (Imron et al. 2020; Kurniawan et al. 2021). For instance, *E. crassipes* can produce up to 45–60 tons of dry biomass per hectare annually under optimal conditions (Malik 2007; Ayanda et al. 2020).

Despite its potential, a significant portion of the harvested biomass is currently discarded into nearby landfills or left to decay, contributing to methane emissions (Wei et al. 2020) and secondary pollution (Ahiati 2021). However, emerging studies have emphasized the economic and environmental value of transforming this biomass into value-added products, including:

- Organic fertilizers and composts, due to the high nitrogen, phosphorus, and potassium content (Singh et al. 2020);
- Bioethanol and biogas, using anaerobic digestion or fermentation (Magdalena et al. 2021);
- Biodiesel, derived from the lipid content of macrophytes such as *Eichhornia crassipes* (Sharma et al. 2021);
- Low-cost adsorbents, prepared from pyrolyzed macrophyte biomass for the removal of heavy metals and dyes (Saralegui et al. 2021);
- Animal feed, particularly after detoxification and proper nutritional analysis (de Queiroz et al. 2020).

The utilization of macrophyte biomass typically follows two major technological pathways:

1. Biochemical methods, such as enzymatic hydrolysis, fermentation, and anaerobic digestion;
2. Thermochemical methods, including pyrolysis, combustion, torrefaction, carbonization, and gasification (Alam et al. 2021).

The yield and quality of end products depend on species type, moisture content, cellulose-to-lignin ratio, and applied technology. For instance, pyrolyzed biomass from *Typha angustifolia* has shown adsorption capacities of up to 78 mg/g for Pb(II), making it a competitive alternative to commercial activated carbon (Yang et al. 2021).

In addition to chemical valorization, structural applications of macrophyte biomass have been explored. Dried and woven stems of macrophytes like *Typha* and *Phragmites* are mechanically strong and can serve as low-cost, biodegradable alternatives for composite building materials and insulation panels, especially in rural and flood-prone areas (Senghor et al. 2024).

Hence, the shift toward circular utilization of macrophyte biomass not only enhances the sustainability of CW systems but also offers solutions for renewable energy production, environmental remediation, and rural livelihood development.

Development of Biochar

Biochar developed from macrophytes biomass has emerged as a promising agent in the reduction of greenhouse gases from the atmosphere and in recovering soil quality (Oni et al. 2019; Antonangelo et al. 2021; Kwoczynski and Cmelik 2021). It has been stated previously that several aquatic macrophytes have the potential to produce biochar with yields extending from 120 to 420 mg /g plant biomass. It can be generated via biomass conversion through thermochemical processes such as gasification, pyrolysis, and torrefaction. Pyrolysis is the main employed process in biochar development (Tripathi et al. 2016; Nand et al. 2024a). The plant biomass can be transformed into biochar since its shoots have enough quantity of carbon, as exhibited earlier by Das et al. (2021). Song et al. (2019) utilized *E. crassipes* biomass to prepare biochar via pyrolysis at a higher temperature (600 °C). They observed that the per gram of biomass produced, 0.03 g biochar was produced and exhibited greater adsorption potential with a maximum adsorption ability (qmax- 0.2 mg/g) for the ammonium nitrogen from piggery farm wastewater. Further, the biomass of various macrophytes such as *Phragmites australis*, *Potamogeton crispus* and *malaianus* and *Vallisneria natans* has been utilized to produce biochar through pyrolysis at 350–600 °C by Xu et al. (2020). They reported that the per gram biomass of these macrophytes has gained the yield extending from 100 to 1000 mg biochar, and also suggested that the yield is significantly enhanced with a rise in temperature. Liu et al. (2019), also produce biochar from biomass of macrophytes such as *Myriophyllum verticillatum L.*, *Ceratophyllum demersum L.*, *Oenanthe javanica* and *Hydrocotyle Vulgaris*, ranging from 360 to 460 mg. It is stated that the rise in the pyrolysis time meaningfully enhances the yield of biochar. Biochar may also be developed via common aquatic plants during eutrophication such as *E. crassipes*, *Phragmites*

australis, *Enteromorpha intestinalis*, *Nelumbo nucifera* and *Hydrocharis sp.* to resolve the handling issue of abundant macrophytes biomass (Sun et al. 2017).

Development of bio-adsorbent

Adsorption is considered an effective method for the treatment of several types of wastewater (Tan et al. 2018; Pullumbi et al. 2019; Nand et al. 2024b). Nevertheless, it is expensive, thus making low-cost alternative adsorbents with inexpensive materials or through waste might be crucial for their applicability (Hynes et al. 2020; Sher et al. 2020). From the literature, it is evaluated that bio-adsorbents may be developed from macrophytic biomass via physicochemical activation procedures and pyrolysis (Qamar et al. 2020; Ani et al. 2020; Reza et al. 2020). Bio-adsorbent has been produced from the biomass of *Scirpus sp.* obtained from the CWs, having greater efficacy for methylene blue removal as explained in a previous study (Alshekhli et al. 2020). Kua et al. (2020) developed an adsorbent from the *Ipomoea aquatica* biomass and achieved up to 76% removal of methyl violet 2B (qmax of 267.9 mg/g). It has been reported that the biomass *Myriophyllum spicatum* is very effective in the removal of several heavy metals such as Zn (II), Pb (II) and Cu (II) (Keskinkan et al. 2003). Likewise, the biomass of *Ceratophyllum demersum* also has huge potential for the removal of Cu (II), Zn (II) and Pb (II) (Keskinkan et al. 2004).

Conversion into biofuel

The production of biofuel from biomass has been extensively studied to develop other sustainable fuels (Vohra et al. 2014). Biofuels, i.e., developed from plant biomass, may be divided into four categories, viz., biodiesel, bioethanol, biogas, and biohydrogen (Rezania et al. 2020; Nand et al. 2024a, b). Bioethanol is mostly manufactured from sugarcane, corn, and other starch-comprising plant biomass (Zabed et al. 2016). The United States was recognized as the top producer of bioethanol (15.78 billion gallons) in 2019, and the second most in Brazil (8.57 billion gallons), as reported earlier by Aruwajoye et al. (2020). Further, the biomass of macrophytes such as *Pistia stratiotes* and *Eichhornia crassipes* was utilized to produce bioethanol, and the yield was recorded up to 0.14–0.17 g/g of biomass (Soda et al. 2013). Miranda et al. (2016) reported that the biomass of *Azolla filiculoides* also produces a great yield of bioethanol (0.09 g). The biomass of various other aquatic plants that are commonly used in CWs, such as *Alternanthera sessilis*, *Ipomoea aquatica*, *Bacopa monnieri*, and *Typha latifolia*, has huge potential for bioethanol generation (Sunil et al. 2015). Similar to bioethanol, biohydrogen is also a substitute fuel that is explored worldwide due to its capability, sustainability, low energy necessity, and high efficacy (Saratale et al. 2019). The majority of biohydrogen produced from plants is associated with the feeding of glucose existing within their biomass via gasification, microbial electrolysis, and dark fermentation (Acar and Dincer 2014; Mthethwa et al. 2019). For example, Miranda et al. (2016) produced 2.2 mol hydrogen/mol glucose from *Azolla sp.* through fermentation. Plants such as *Landoltia punctata* and *Azolla filiculoides* were used to produce biohydrogen and recorded the yield of 1.8 mol hydrogen (Miranda et al. 2020). Further, the biomass of *Pistia stratiotes* as feedstock was reported to produce biohydrogen upto 2.2 mol hydrogen/mol glucose via fermentation (Mthethwa et al. 2019).

Further, biogas may be utilized as fuel and fuel cells in the generation of electricity and running microturbines however, the remaining digestate may be used as biofertilizer (Ani et al. 2020). It is produced from the feedstock of cattle dung, and biomass of municipal, agricultural and other lignocellulosic waste including plants containing a high proportion of cellulose, fats, hemicellulose, carbohydrates and proteins (Ghosh et al. 2020). However, the generation of biogas is extremely reliant on several other factors, like feedstock type, retention time and digestion system, etc. (Ghosh et al. 2020). The biomass of macrophytes is employed in anaerobic digesters to generate biogas (0.89 to 5 mL biogas/g biomass) as reported earlier (Moeller et al. 2018). The biomass of several macrophytic species, specifically, *Canna indica*, *Typha orientalis*, *Acorus calamus*, *Eichhornia crassipes*, *Thalia dealbata*, *Hydrocotyle Vulgaris*, *Colocasia tonoimo*, and *Pontederia cordata*, is applied as substrates in the generation of biogas via anaerobic digestion. Jiang et al. (2014), obtained 2.75 mL biogas/g of biomass of *Thalia dealbata* and *Hydrocotyle vulgaris*. However, Almoustapha et al. (2009) recorded 3 to 5 mL biogas/g of biomass of *Eichhornia crassipes* in the summer and winter seasons, respectively. Further, the biomass of aquatic plants such as *Azolla filiculoides* has also been exploited as raw materials for the generation of biodiesel. A previous laboratory-scale study reports that the annual biodiesel production with *Azolla filiculoides* could theoretically achieve 1.68 t/ha/year as compared to Glycine max (0.44 t/ha/year) (Miranda et al. 2016). Salehzadeh et al. (2014), generated biodiesel with the same plant and gained up to 150 mg biodiesel/g of biomass. Moreover, biodiesel may also be generated in the range of 32.2–63.6 mg/g biomass of *Eichhornia crassipes* as a raw material (Shanab et al. 2018).

Development of organic fertilizer

The biomass of several aquatic plants can be exploited as a raw resource to develop various types of organic fertilizers.

However, the production of fertilizers depends extremely on the efficacy of the digestion system (Stabenau et al. 2018). Several studies have pointed out that harvested *Phragmites*, *Typha*, and other macrophytes from CWs are co-composted to produce the biofertilizer. The application of co-composting transforms complex organics into simple compounds and steady humus (Torres-Climent et al. 2015; Diacono et al. 2019). Stabenau et al. (2018), stated that the yield of phosphorus fertilizer was recorded in the range of 0.5–6.3 mg P/g biomass of *Elodea nuttallii* after digestion. The produced solid digestates chiefly comprise nitrogen (20%–25%), organic matter, and phosphorus (60%–80%) and potassium (10%–15%) (Möller and Müller 2012). However, pretreatment of digestate is recommended to gain valued biofertilizers as their direct application is inadequate due to odor, phytotoxicity, and viscosity (Kurniawan et al. 2021). The method may be upgraded through the introduction of co-composting, vital for the recycling of organic matter and agricultural production, and to enhance microbial activities.

Future concerns and challenges

CWs have an extensive range of applications appropriate for the future employment of a resource-oriented circular economy and ecosystem amenities paradigm (Stefanakis et al. 2021). Nevertheless, these innovative claims need further research and explanation for real use. Further, the ecosystem-based approaches should be incorporated into every urban planning, introducing multi-purpose blue and green infrastructures in cities (Liquete et al. 2016; Masi et al. 2018). The plants that are utilized in CWs for the treatment of highly contaminated wastewater particularly with dyes, endocrine disrupting compounds, emerging contaminants (Ahmed et al. 2021) and/or pharmaceuticals, limit their utilization in fertilizers production and animal feedstock. These compounds may not be degraded and can build up in cell tissues and distress the food chain. Thus, the selection of the right plant biomass as feedstock should be examined in terms of their nutritional values for animals' well-being (Kurniawan et al. 2021). Further, the selection of macrophytes for contaminant uptake efficiency and providing and supporting the growth and adherence of microbial populations is also crucial. CWs require large land areas for their effective operation however, this can be tackled through hydroponic or staged systems. The seasonal variations and existing environmental settings may also hinder the growth of macrophytes and the activity of microorganisms (Al-Baldawi et al. 2021). To overcome these challenges, aeration, bioaugmentation of rhizobacteria and supply of enzymes might be good options to enhance the removal rate (Kumar et al. 2022a, b). Further, the choice of suitable technology is one of the critical concerns confronted by authorities to effectively implement circular economy strategies of recycling, reclamation and recovery (Stefanakis et al. 2021).

Lack of expertise or trained manpower, policies encouraging the use of recovered bioresources. The benefits or incentives of utilizing recovered resources are also not well-defined, thus, affecting the effective working of the system and implementation of the circular economy initiative (Nikolaou and Stefanakis 2022). During the application of CWs, variable qualities of plant biomass are produced due to differences in wastewater quality, leading to a varying yield of biogas as well as biofuels. To overcome these challenges, there is a need for multidisciplinary studies (Kurniawan et al. 2021). The implementation of the circular economy in existing infrastructure needs the collaboration of researchers and policymakers of different disciplines such as natural sciences, social sciences and engineering, etc. (Fig. 4). Governments should also encourage industries to implement circular strategies in their commercial goals by offering tax reductions/exemption incentives (Korhonen et al. 2018; Kakwani and Kalbar 2020).

Conclusion

It is well explained that several dyes existing in textile effluent are considered genotoxic, mutagenic and carcinogenic for all life forms. Various physicochemical and biological methods are employed in the treatment of textile wastewater such as filtration, flocculation, coagulation, adsorption, anaerobic digestion, and chemical oxidation and bioreactors, etc. However, all these technologies are considered expensive and generate secondary pollutants. CWs have emerged as excellent alternatives for the effective treatment of textile wastewater together with resource recovery options for the circular economy. It is explained as an appropriate and efficient economic model that mainly focuses on the use of resources sensibly and enhances their efficiency while reducing the generation of waste, emission, energy consumption and harmful impact on the ecosystem. The biomass of several plants can be exploited as a raw material to develop different bioresources such as biofertilizers, biofuels, bioenergy, animal feed, biochar, recovery of various nutrients and pharmaceutical compounds, etc. Biochemical and thermochemical technologies are usually employed to process macrophytic biomass. The biochemical approach comprises of extraction and hydrolysis and thermochemical methods include pyrolysis, combustion, activation and gasification. However, the end products differ greatly because of the different characteristics of the plants used, the purpose of the product used and the technology employed. In

Fig. 4 The implementation of new R's in the circular economy concept for further advancements

addition, strong, robust and air-dried strands of the stem of several plants may be utilized as alternative structural resources which can be used as a substitute for expensive and corrosive materials. Resultant sludge and other solid, biosolids and semi-solid waste may be utilized to produce heat or electricity, biogas, and transform them into organic manure or fertilizers.

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